The Complex Case of Impact Reporting in an NPO

A Design Science Research Approach

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01

Motivation & research question

Digital technologies promise faster and better ways for NPOs to highlight impact and communicate to funders and beneficiaries.

Digital technologies promise faster and better ways of highlighting impact and communicating to funders and beneficiaries.

But meaningful automation of impact reporting is no easy task.

Digital era: pressure for quantitative data to showcase impact.

Impact reporting can overshadow social impact

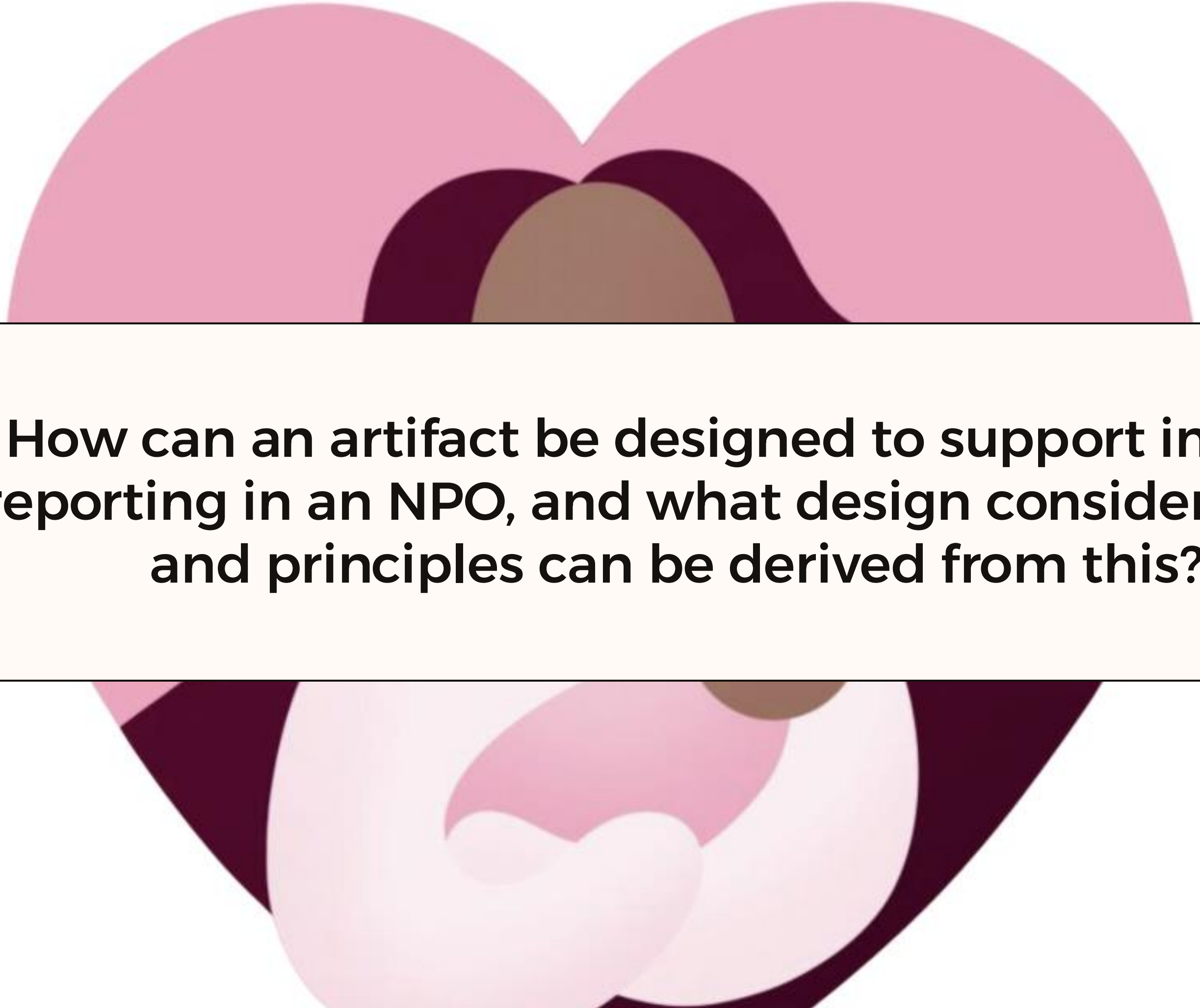
Value-creation structure: Social commitment, but dependent on funding

Digital technologies promise faster and better ways of highlighting impact and communicating to funders and beneficiaries.

Lack of prescriptive guiding principles

How to meaningfully automate impact reporting in NPOs?

This illustrate the relevance and motivation of our project.



How can an artifact be designed to support impact reporting in an NPO, and what design considerations and principles can be derived from this?

02

DSR-approach & Empirical Grounding

Why Design Science Research?

Why Design Science Research?

DSR: Problem-solving paradigm that address real-world problems through artifacts

Investigate a real-world problem

Design an artifact – unforeseen challenges appear

Bridging theory and practice – translating insights to prescriptive principles

Combining Hevner, Peffers & Venable in our DSR approach

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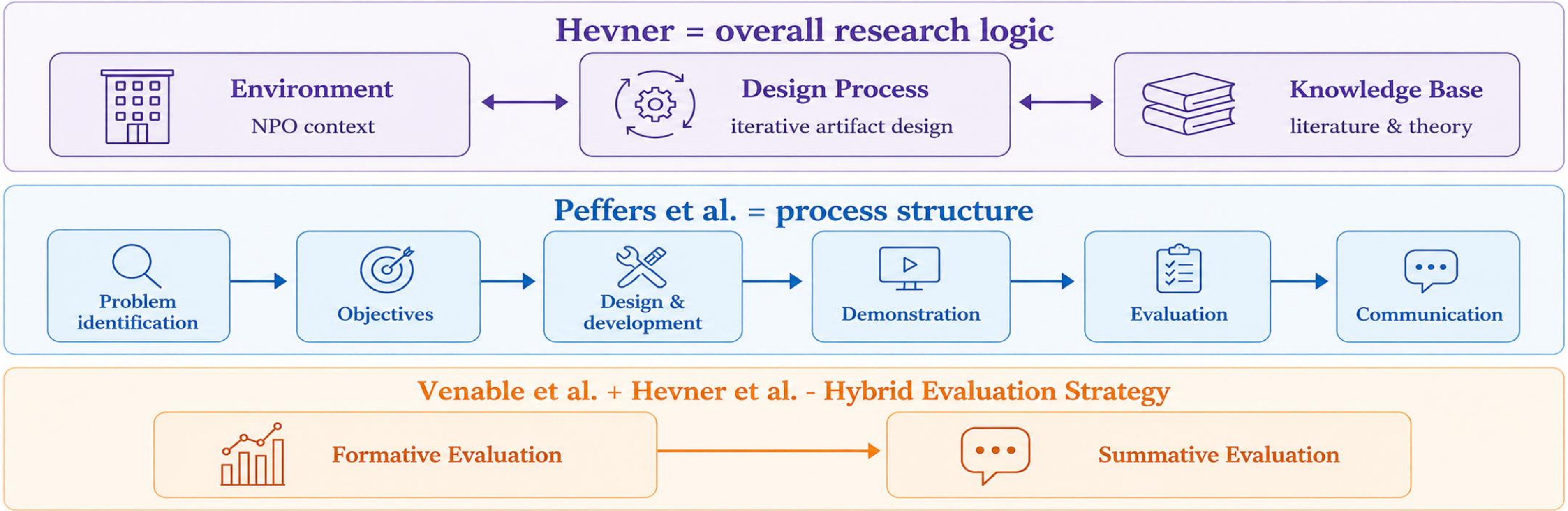
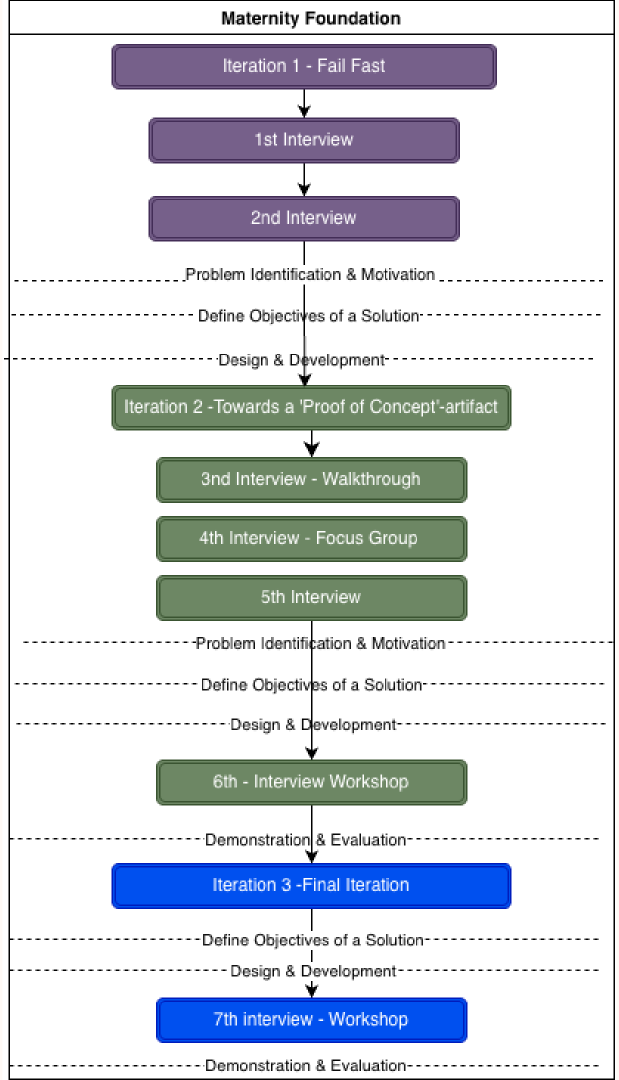
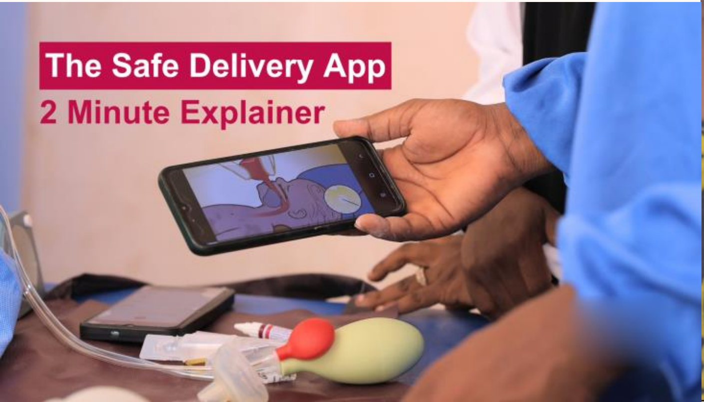


Sketching out our DSR-approach: The Three Iterations



Maternity Foundation

- Partners in +70 LMIC countries
- 2015: Safe Delivery App
- +500.000 midwives and healthcare professionals



maternity
FOUNDATION

MATERNITY FOUNDATION

ORGANISATIONAL STRUCTURE

INSIGHT, RESEARCH AND LEARNING DEPARTMENT (IRL)

Works with collecting, processing, analyzing, and visualising data for clinical trainings, app development, internal processes, fundraising, impact reporting etc.

Also responsible for writing yearly-, grant-, and impact reports.

HEAD
Atana

TEAM MEMBERS
Ileen
Asha

IT DEPARTMENT

Takes care of all technological, infrastructural and digital aspects of the organisation.

Any changes to MF's infrastructure needs to be approved by them as they are very concerned with containing the monetary costs.

HEAD
Tian

PROGRAM DEVELOPMENT DEPARTMENT

Responsible for scaling the Safe Delivery APP, strengthen health systems and expand the network with partners and collaborators.

TEAM MEMBER
Ulrika

BUSINESS DEVELOPMENT DEPARTMENT

Responsible for fundraising, grant management and impact reporting.

TEAM MEMBER
Yvette

CLINICAL DEPARTMENT

Responsible for preparing, executing and developing clinical trainings.

They travel to clinical trainings to train local healthcare professionals.

TEAM MEMBER
Eke

COMMUNICATIONS DEPARTMENT

Responsible for MF's national and international communication activities, with a core focus on visual communication and branding.

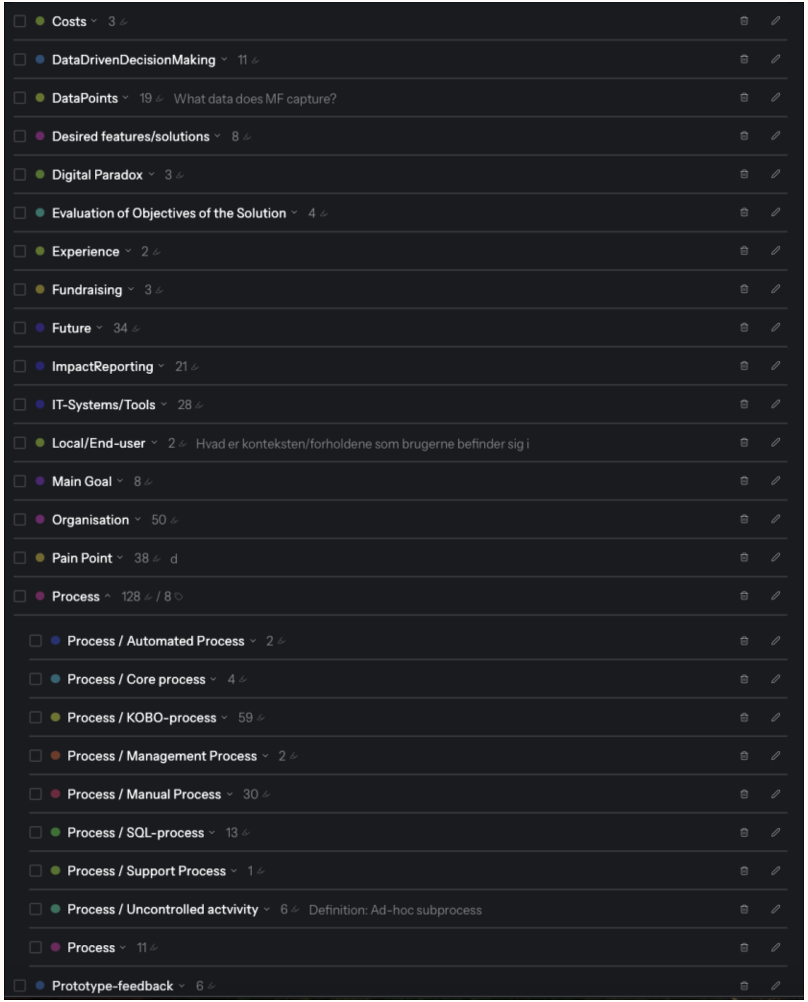
TEAM MEMBER
Eisha

Grounded in Empirical Data

Stakeholder's descriptions helped shape and define the problem (empirically grounded)

To organize and structure the interviews, we conducted a **thematic analysis (Braun & Clarke)**

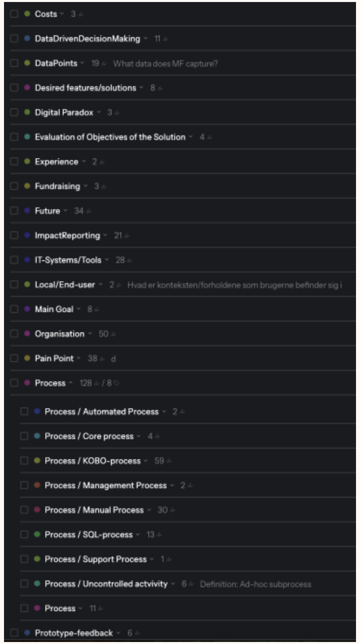
Via the software-tool ***EthNote.***



Grounded in Empirical Data

Highly iterative process between
1) **Collecting data**
2) **Understanding the problem/phenomena**
3) **Engaging with literature**
4) **Designing an artifact**

Phase 2 – **generating 28 initial codes**



Phase 3 – Combining 28 codes to **8 overarching themes**

Theme
Lack of standardization
Lack of strategic alignment
Resource constraints
IT-set up
Organizational purpose and goals
Barriers to digital transformation
Accountability and impact reporting
5-year futuristic plan

Phase 4–5 – Revised and combined into **5 final themes**

Reviewing Themes
Standardization & automation
Strategy for digital transformation
Cultural resistance towards standardization
Social impact vs. impact reporting
Local IT-infrastructure and IT-skills

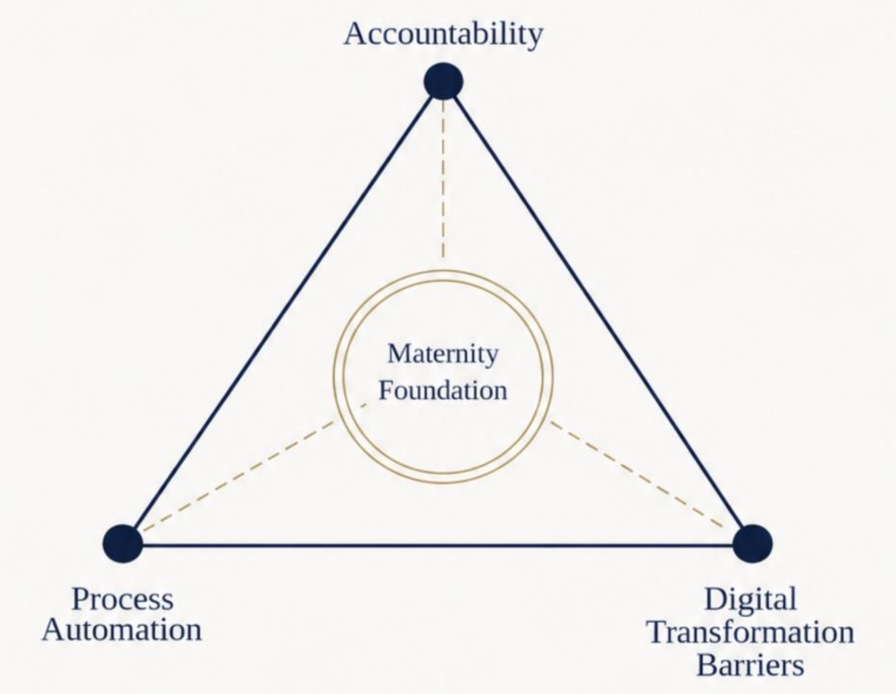
03

Theoretical Perspective

Theoretical Perspective
To inform our DSR-approach & Artifact

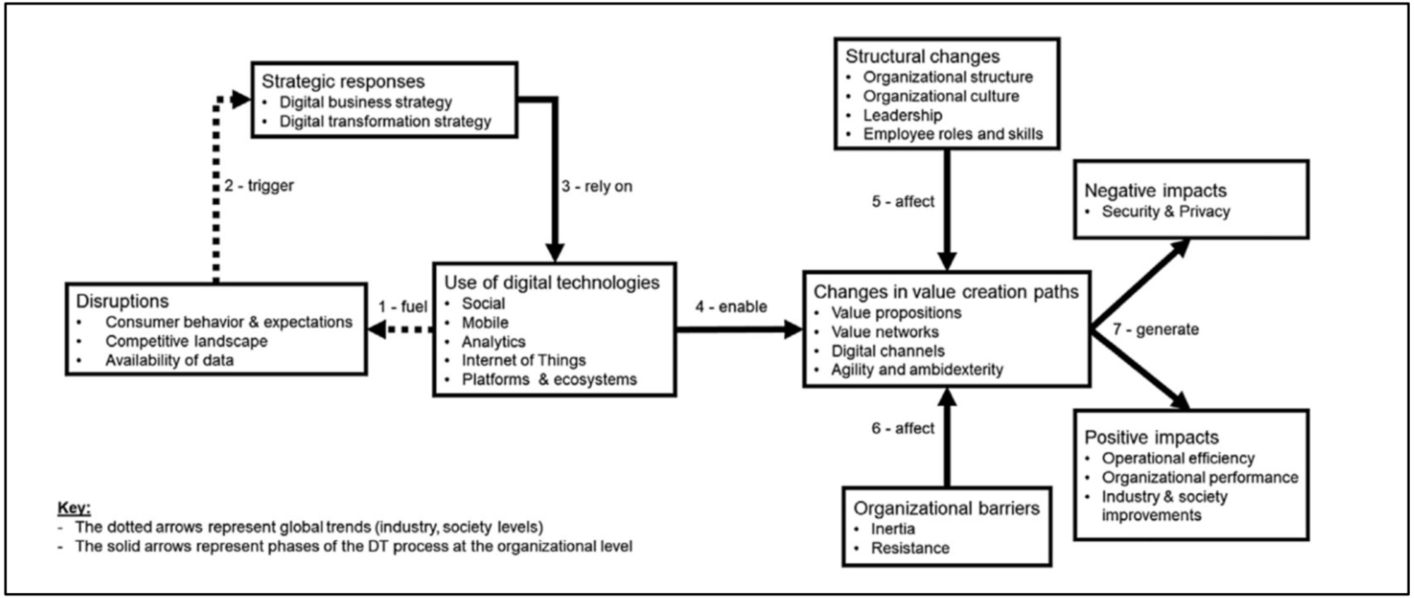
3 Concepts:

- (1) Digital Transformation Processes
- (2) Process Automation and RPA
- (3) Accountability in NPOs



Theoretical Perspective
Digital Transformation Processes

- Captures **multidimensional** and **transformative** aspects of **implementing** digital technologies
  - The guiding framework for theoretically conceptualizing our empirical findings.



Theoretical Perspective
Situating Digital Transformation in NPO

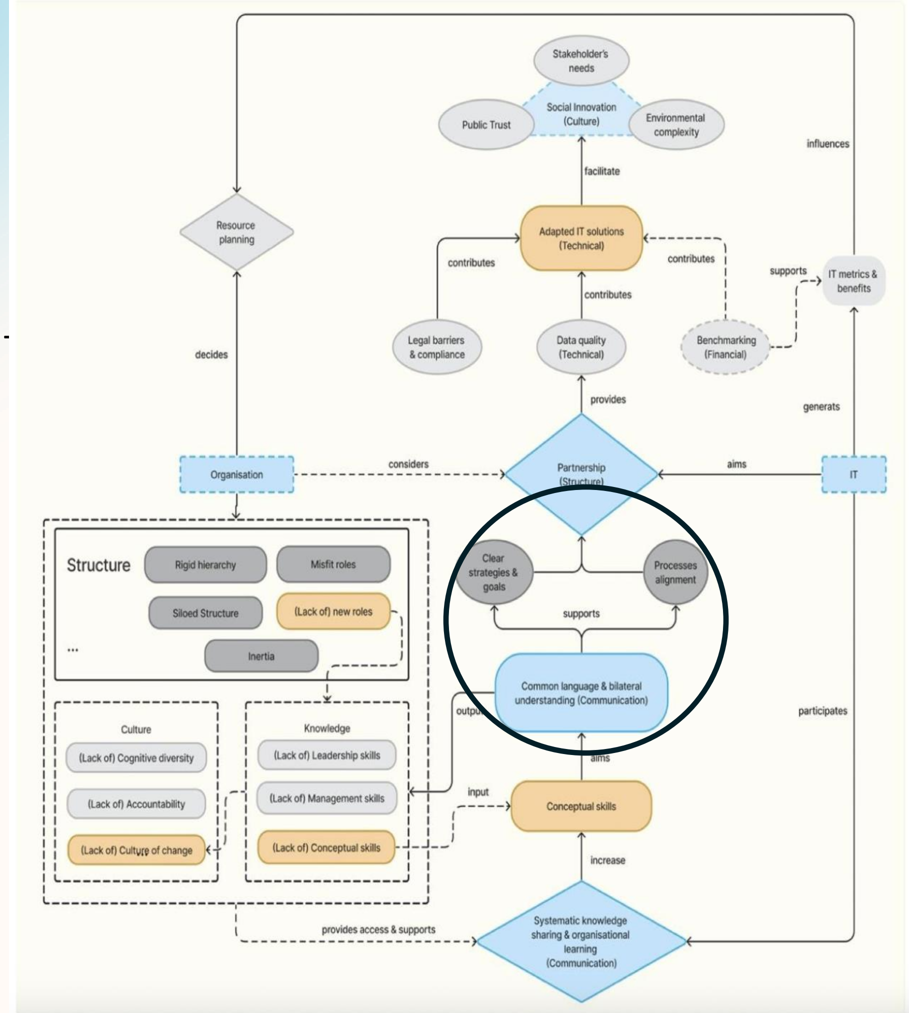
Accountability – and the **performative calculative practices.**

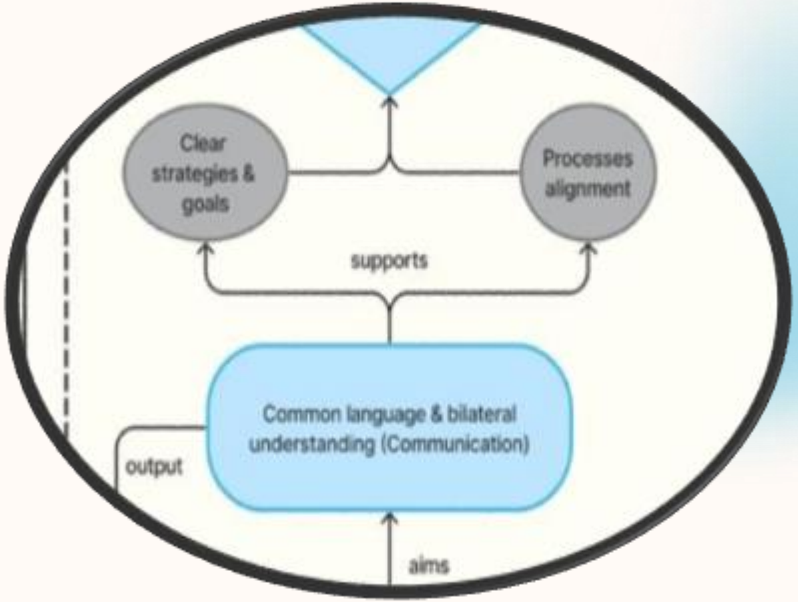
- Risk that **efficiency** and **cost control** overshadows **social mission.**

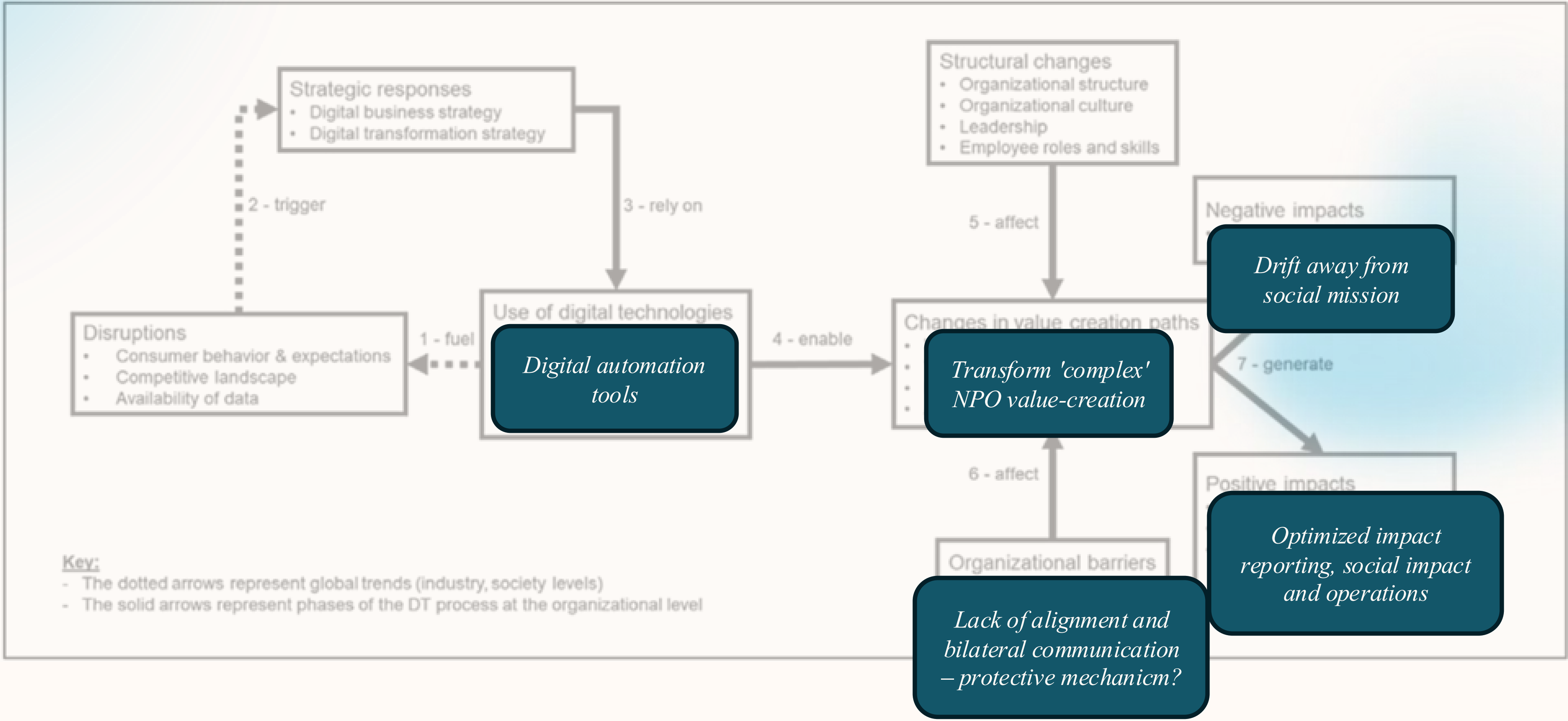
"Peculiar" value-creation.

- Committed to **social mission,** but **dependent** on **external funding**

Process automation & RPA as the technological entry point.


Stakeholder's needs
Social Innovation (Culture)
Public Trust
Environmental complexity
influences
facilitate
Resource planning
Adapted IT solutions (Technical)
contributes
contributes
contributes
supports
IT metrics & benefits
Legal barriers & compliance
Data quality (Technical)
Benchmarking (Financial)
decides
provides
generats
Organisation
considers
Partnership (Structure)
aims
IT
Structure
Rigid hierarchy
Misfit roles
Siloed Structure
(Lack of) new roles
...
Inertia
Clear strategies & goals
Processes alignment
supports
Common language & bilateral understanding (Communication)
Culture
(Lack of) Cognitive diversity
(Lack of) Accountability
(Lack of) Culture of change
Knowledge
(Lack of) Leadership skills
(Lack of) Management skills
(Lack of) Conceptual skills
participates
aims
input
Conceptual skills
increase
provides access & supports
Systematic knowledge sharing & organisational learning (Communication)


Clear strategies & goals
Processes alignment
supports
Common language & bilateral understanding (Communication)
output
aims


Strategic responses
Digital business strategy
Digital transformation strategy
2 - trigger
3 - rely on
Structural changes
Organizational structure
Organizational culture
Leadership
Employee roles and skills
5 - affect
Negative impacts
Drift away from social mission
Disruptions
Consumer behavior & expectations
Competitive landscape
Availability of data
1 - fuel
Use of digital technologies
Digital automation tools
4 - enable
Changes in value creation paths
Transform 'complex' NPO value-creation
7 - generate
6 - affect
Positive impacts
Optimized impact reporting, social impact and operations
Organizational barriers
Lack of alignment and bilateral communication – protective mechanicm?
Key:
- The dotted arrows represent global trends (industry, society levels)
- The solid arrows represent phases of the DT process at the organizational level

04

Problem Identification (First DSR-activity)

Problem Identification
Process selection

Business Process Landscape

Performed to help navigate and understand how the various processes presented in the interviews relate to the core processes of the organization.

Pointless to improve a process that doesn't add value

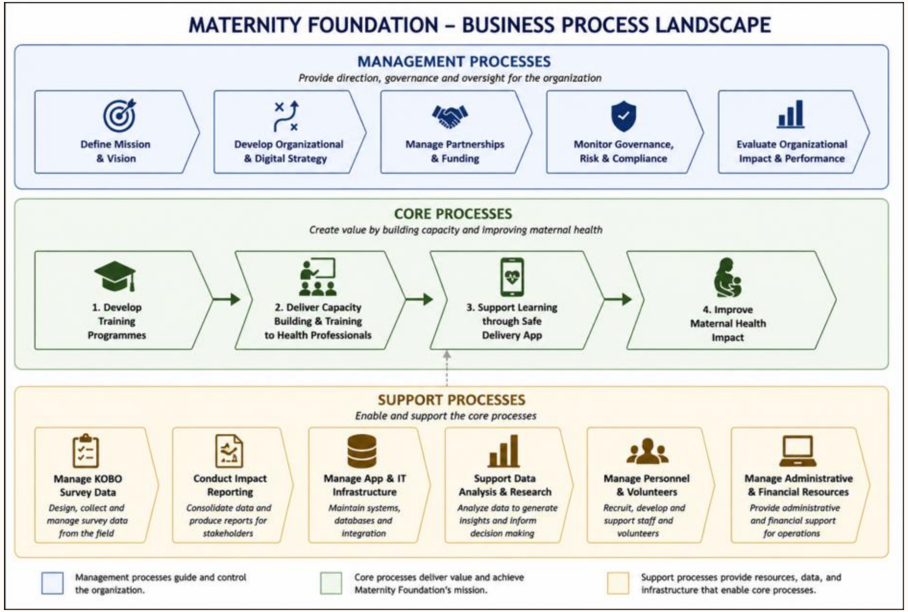
Process Portfolio Matrix

What process is meaningful to improve?

Feasibility + Importance + Health

Problem Identification
Process selection

Business Process Landscape



Process Portfolio Matrix

The Chosen Process: "Conduct Learning & Impact Reporting (Training), *CLIRT-process*"

Problem Identification
Process selection

The Chosen Process: "Conduct Learning & Impact Reporting (Training), *CLIRT-process"*

What does the process consist of?

- *In short:* ***Captures data before and after trainings to showcase the impacts of trainings.***
- *In short:* ***Measured primarily through increase in confidence, knowledge and skills.***

Problem Identification *Formulation*

Iteration 1:

"The selected process – Conduct Learning & Impact Reporting (Training) – is characterized by **extensive manual work** practices, **fragmented data flows**, and **limited integration** between reporting activities and existing app data. Consequently, the current process creates significant **operational inefficiencies** and **constrains** the organization's ability to conduct **scalable impact reporting**."

Technology-oriented

Final Iteration:

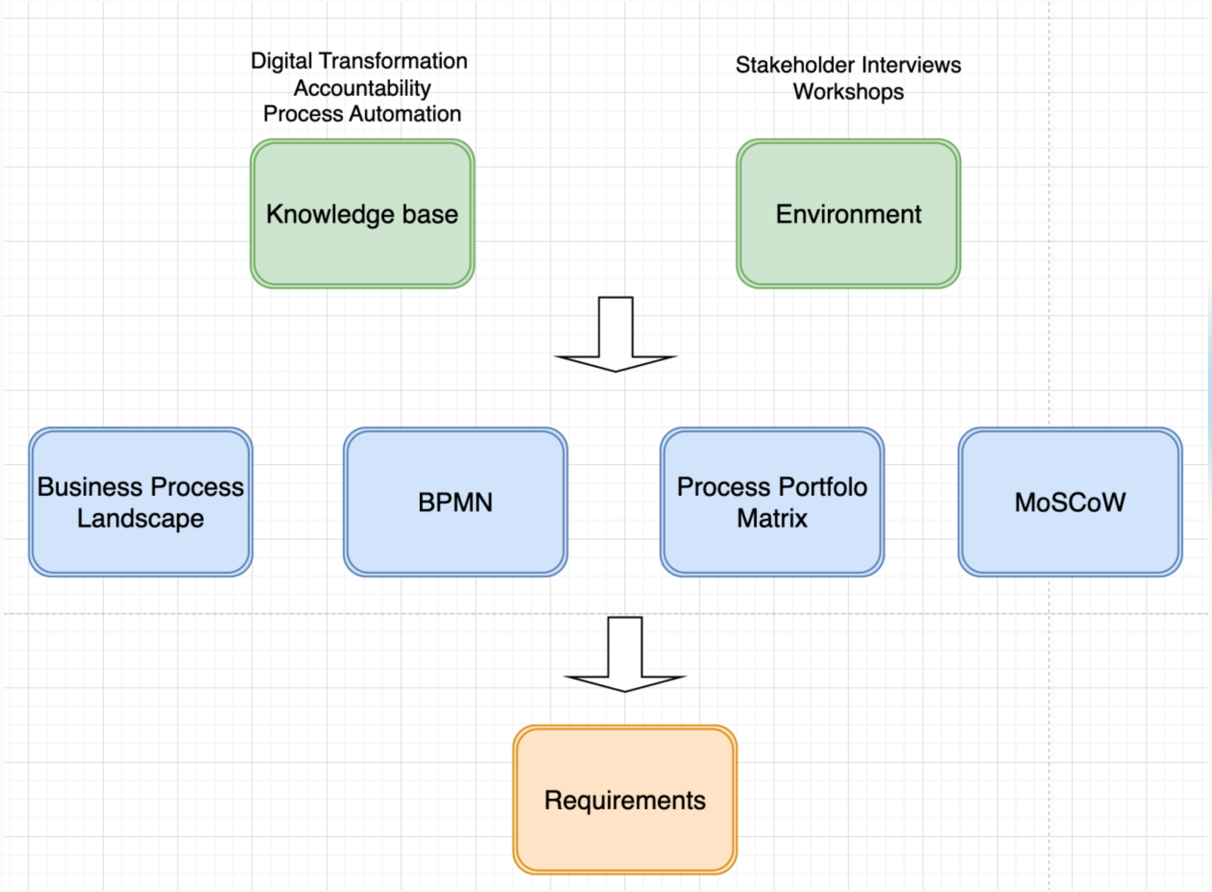
"The organization lacks a ***common language*** and ***bilateral communication*** for critically discussing and assessing ***the trade-offs*** involved in ***standardizing survey data*** to enable ***process automation*** and ***scalability*** compared to the potential loss of flexibility and contextual insights."

Social & organizational oriented

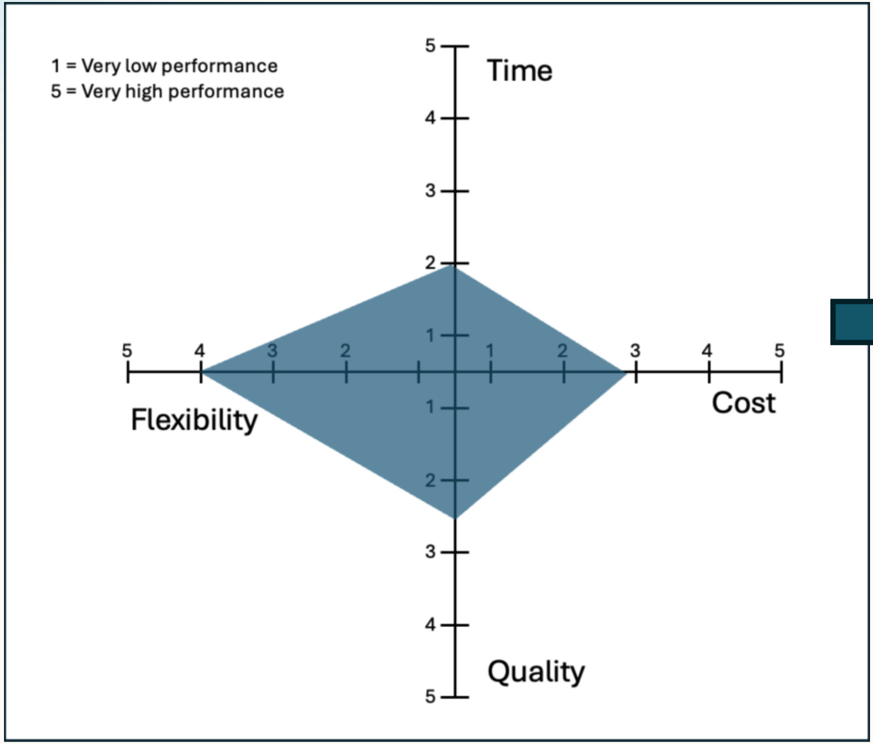
05

Objectives of a solution

Defining objectives of a solution



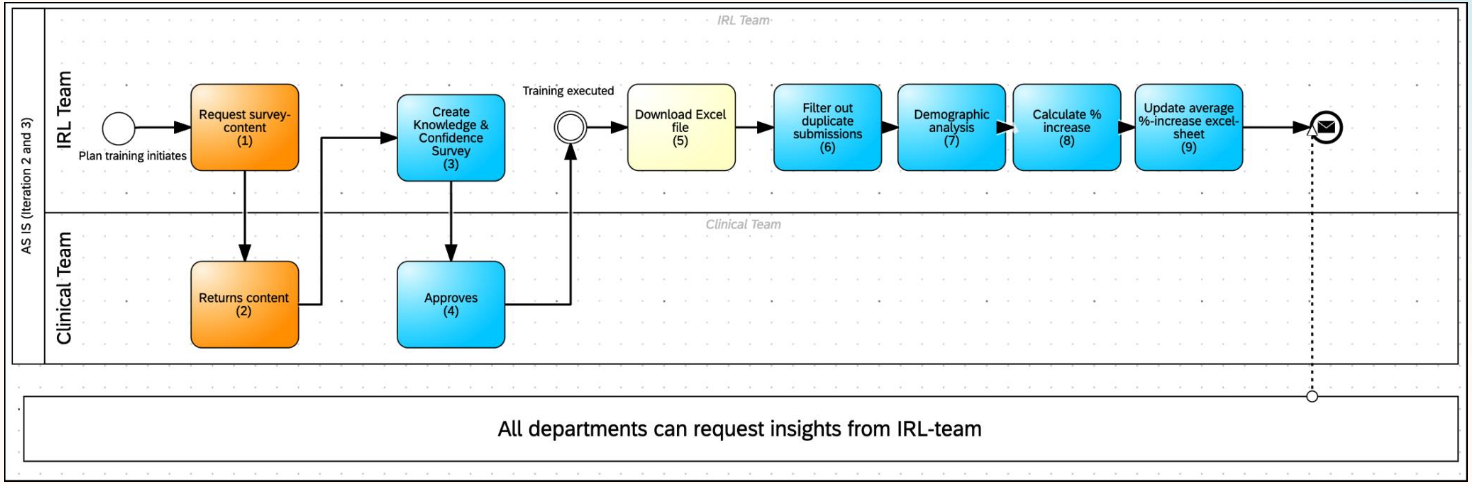
Defining objectives of a solution

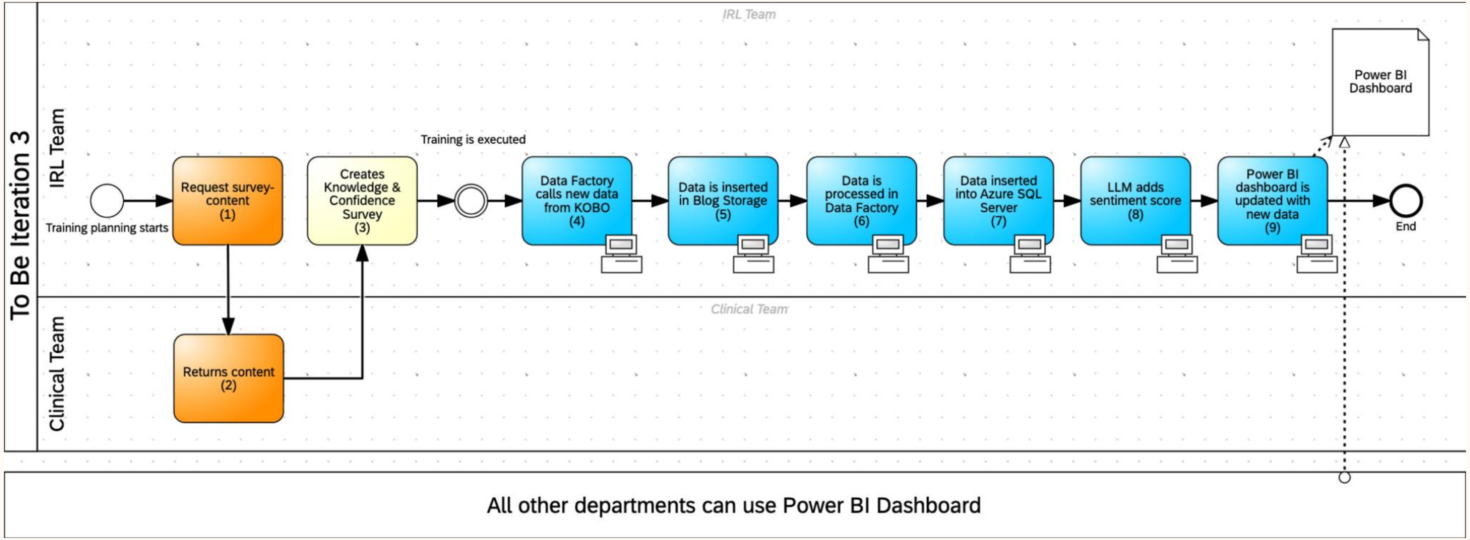


Must	Should
- Support cross-departmental discussion regarding standardization trade-offs - Visualize operational implications of process automation - Include a visual representation of a "KOBO-dashboard" - Use Data Factory as a Middle Layer between KOBO and Power BI - Extract KOBO-data into SQL-server in relational data format	- KOBO-data should be stored in a format allowing comparability with APP-data - KOBO-data should get extracted out of KOBO - Preserve interpretability of local survey contexts
Could	**Won't**
- Support a rethinking of the current KOBO-survey data-infrastructure. - Touch upon the performance dimensions of cost, time and quality - Present our example of a standardized "Master Form"?	- Full organizational implementation/integration - Be categorical and argue for what is the right solution moving forward

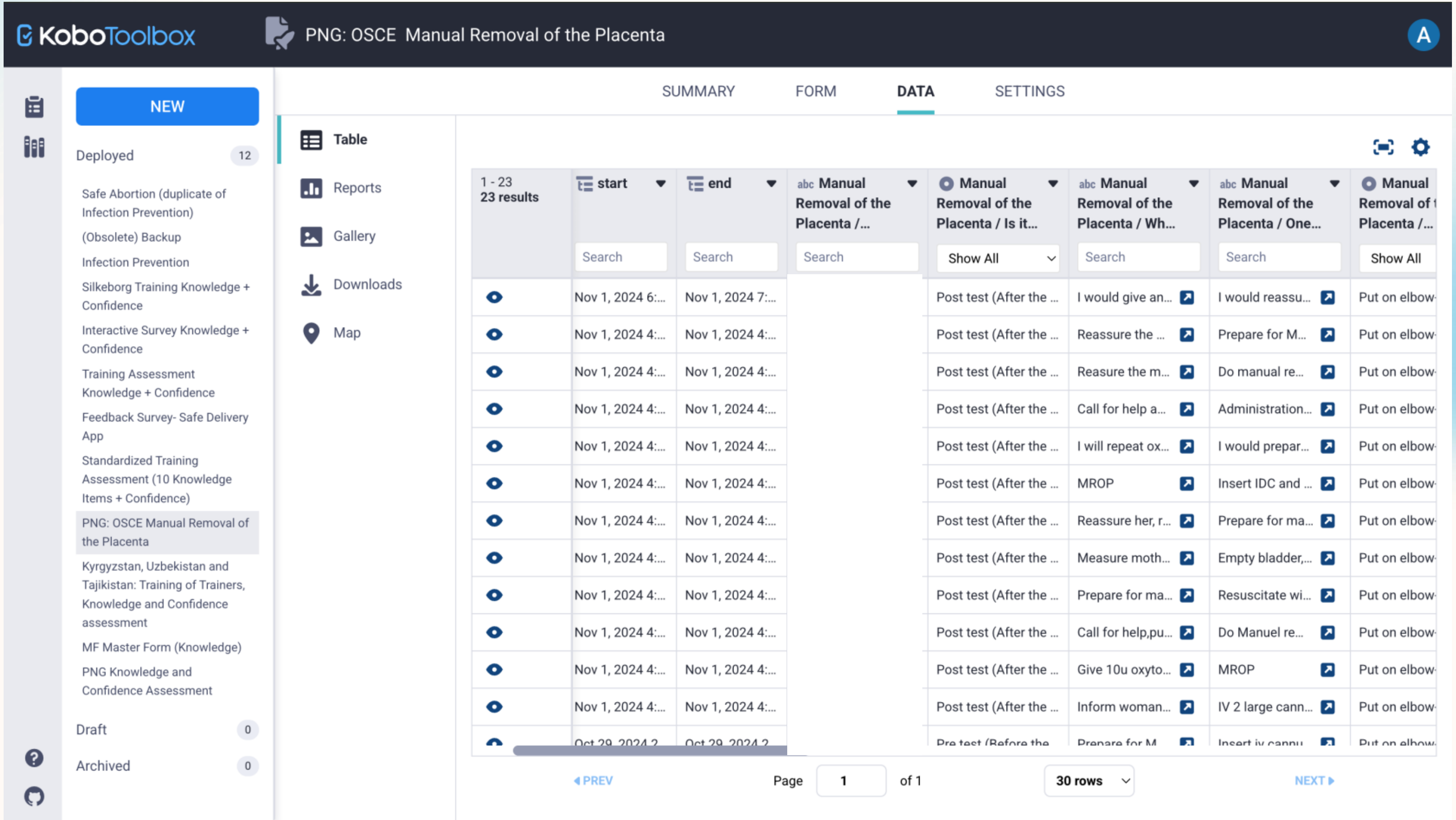
06

Design and Development


AS IS (Iteration 2 and 3)
IRL Team
Clinical Team
Plan training initiates
Request survey-content (1)
Returns content (2)
Create Knowledge & Confidence Survey (3)
Approves (4)
Training executed
Download Excel file (5)
Filter out duplicate submissions (6)
Demographic analysis (7)
Calculate % increase (8)
Update average %-increase excel-sheet (9)
All departments can request insights from IRL-team

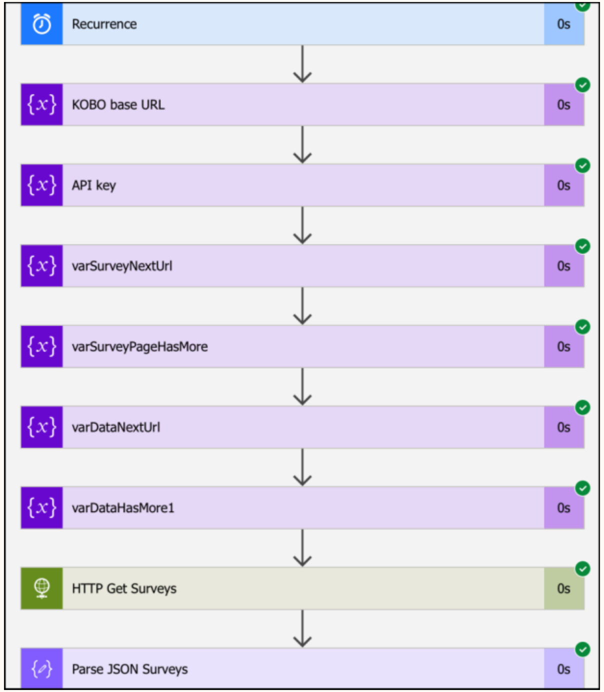

To Be Iteration 3
IRL Team
Clinical Team
Training planning starts
Request survey-content (1)
Returns content (2)
Creates Knowledge & Confidence Survey (3)
Training is executed
Data Factory calls new data from KOBO (4)
Data is inserted in Blog Storage (5)
Data is processed in Data Factory (6)
Data inserted into Azure SQL Server (7)
LLM adds sentiment score (8)
Power BI dashboard is updated with new data (9)
End
Power BI Dashboard
All other departments can use Power BI Dashboard

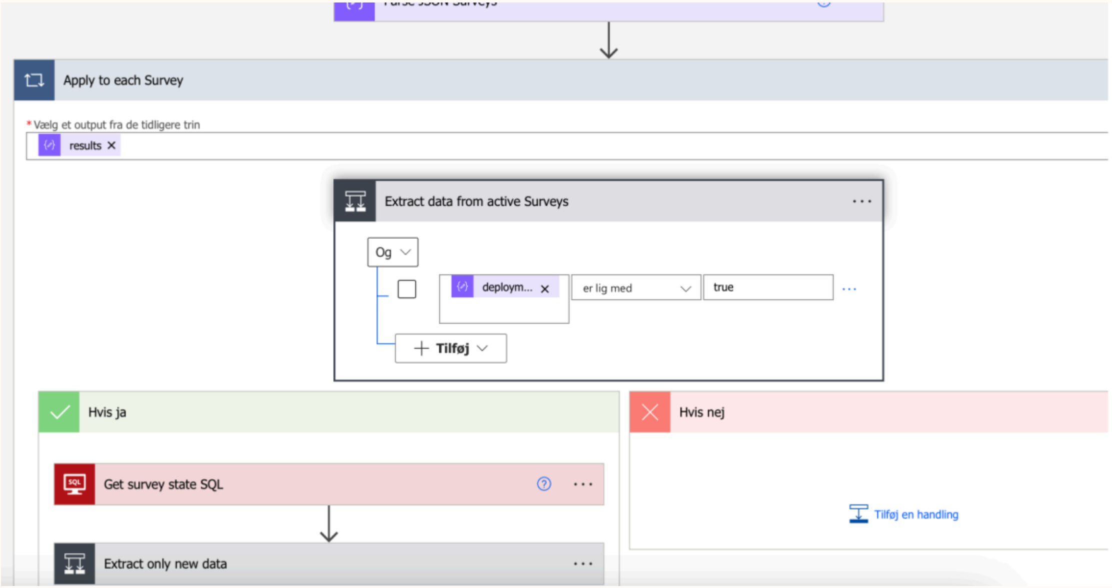
KOBO



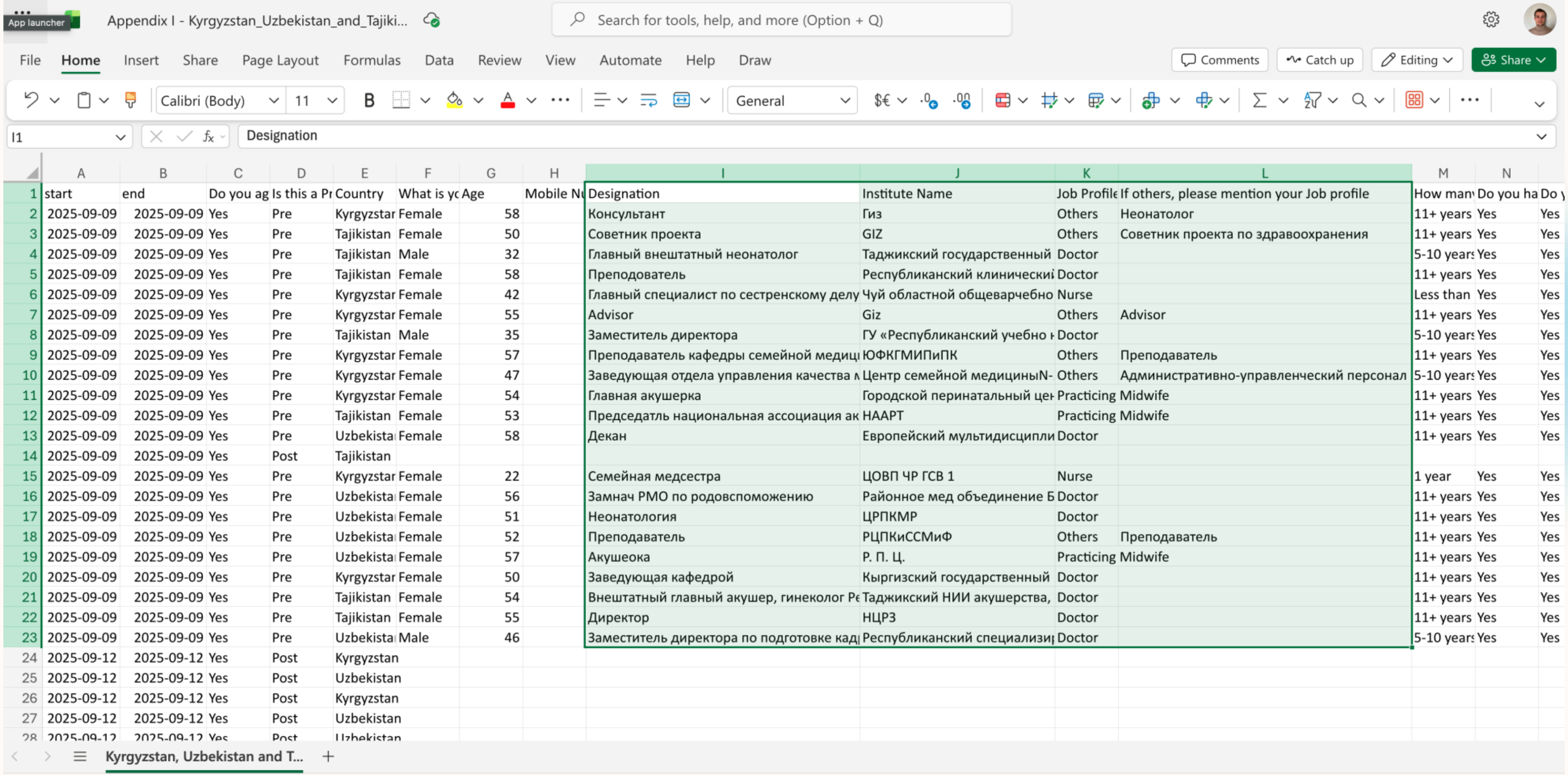
KoboToolbox
PNG: OSCE Manual Removal of the Placenta
A
NEW
Deployed 12
Safe Abortion (duplicate of Infection Prevention)
(Obsolete) Backup
Infection Prevention
Silkeborg Training Knowledge + Confidence
Interactive Survey Knowledge + Confidence
Training Assessment Knowledge + Confidence
Feedback Survey- Safe Delivery App
Standardized Training Assessment (10 Knowledge Items + Confidence)
PNG: OSCE Manual Removal of the Placenta
Kyrgyzstan, Uzbekistan and Tajikistan: Training of Trainers, Knowledge and Confidence assessment
MF Master Form (Knowledge)
PNG Knowledge and Confidence Assessment
Draft 0
Archived 0
SUMMARY
FORM
DATA
SETTINGS
Table
Reports
Gallery
Downloads
Map
1 - 23
23 results
start
end
Manual Removal of the Placenta /...
Manual Removal of the Placenta / Is it...
Manual Removal of the Placenta / Wh...
Manual Removal of the Placenta / One...
Search
Show All
Nov 1, 2024 6:...
Nov 1, 2024 7:...
Nov 1, 2024 4:...
Post test (After the ...
Pre test (Before the
I would give an...
Reassure the ...
Reasure the m...
Call for help a...
I will repeat ox...
MROP
Reassure her, r...
Measure moth...
Prepare for ma...
Call for help,pu...
Give 10u oxyto...
Inform woman...
I would reassu...
Prepare for M...
Do manual re...
Administration...
I would prepar...
Insert IDC and ...
Prepare for ma...
Empty bladder,...
Resuscitate wi...
Do Manuel re...
IV 2 large cann...
Put on elbow
PREV
Page
1
of 1
30 rows
NEXT

1st iteration 'Fail Fast'



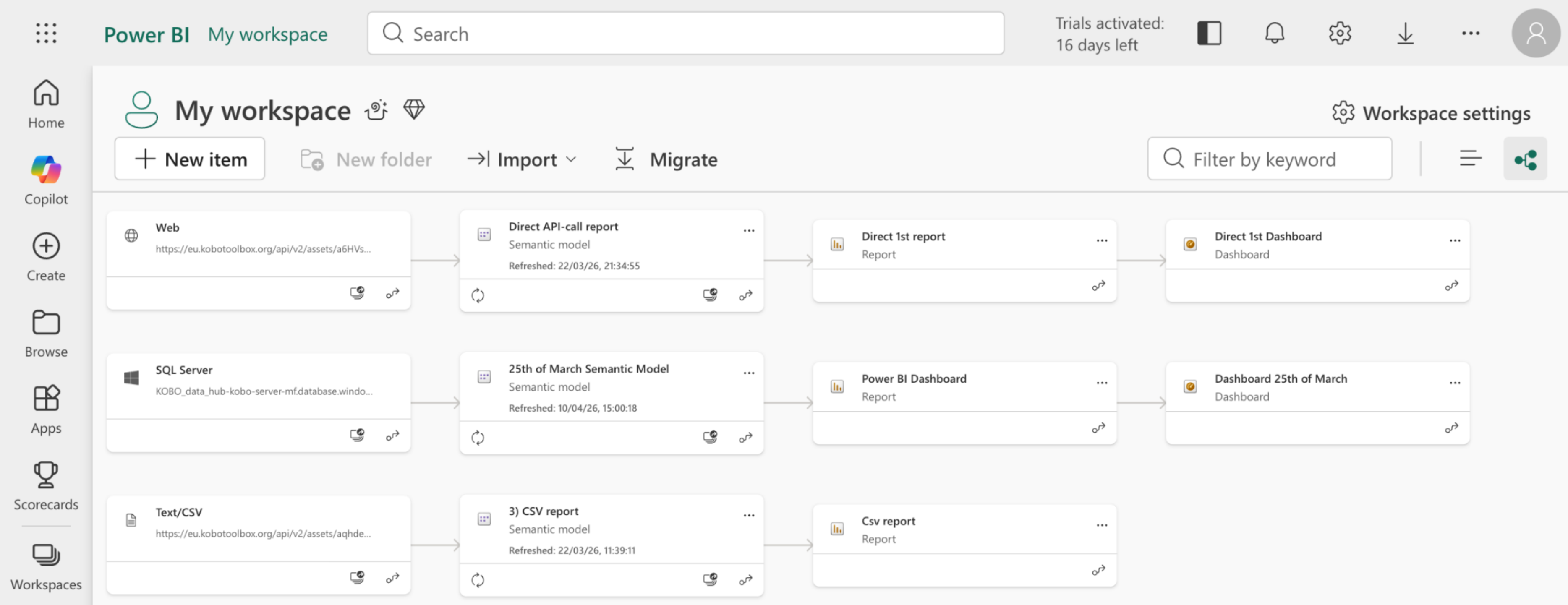


1st iteration 'Fail Fast'

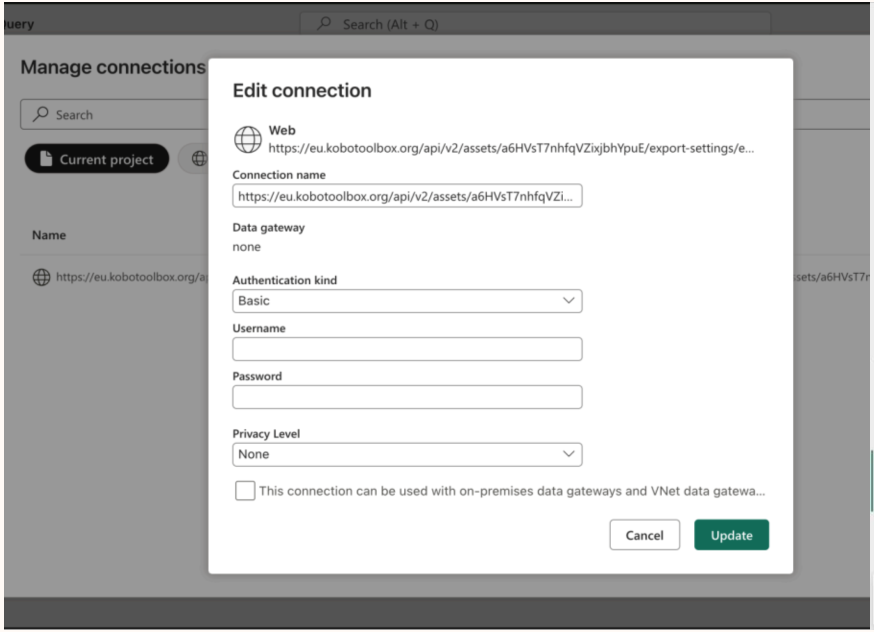


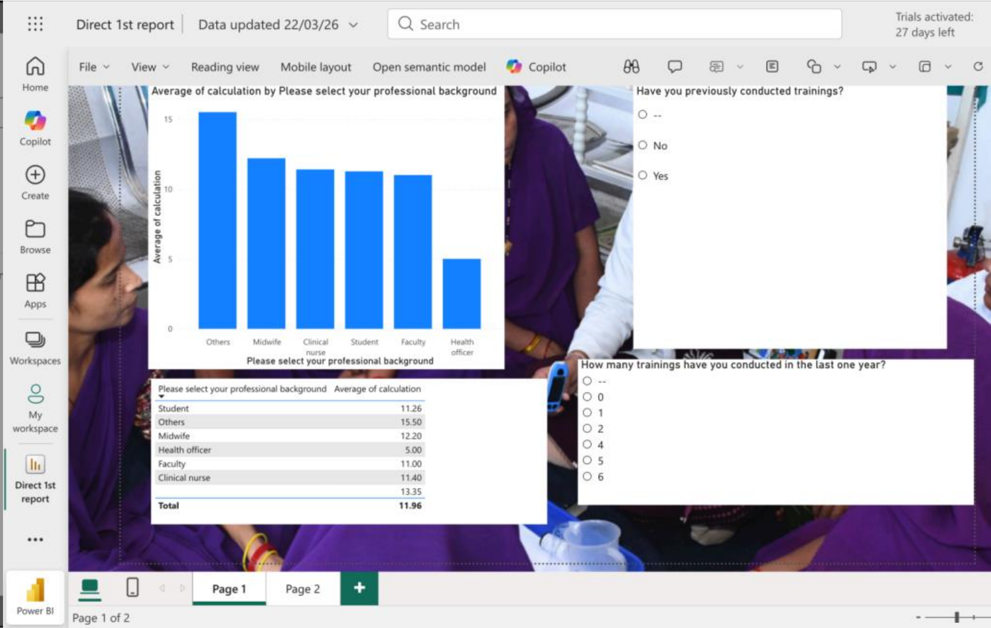
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
1	start	end	Do you ag	Is this a Pr	Country	What is yc	Age	Mobile Nu	Designation	Institute Name	Job Profile	If others, please mention your Job profile	How many	Do you ha	Do y
2	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	58		Консультант	Гиз	Others	Неонатолог	11+ years	Yes	Yes
3	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Female	50		Советник проекта	GIZ	Others	Советник проекта по здравоохранения	11+ years	Yes	Yes
4	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Male	32		Главный внештатный неонатолог	Таджикский государственный	Doctor		5-10 years	Yes	Yes
5	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Female	58		Преподователь	Республиканский клинически	Doctor		11+ years	Yes	Yes
6	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	42		Главный специалист по сестренскому делу	Чуй областной общеварчебно	Nurse		Less than	Yes	Yes
7	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	55		Advisor	Giz	Others	Advisor	11+ years	Yes	Yes
8	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Male	35		Заместитель директора	ГУ «Республиканский учебно	Doctor		5-10 years	Yes	Yes
9	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	57		Преподаватель кафедры семейной медици	ЮФКГМИПиПК	Others	Преподаватель	11+ years	Yes	Yes
10	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	47		Заведующая отдела управления качества	Центр семейной медициныN-	Others	Административно-управленческий персонал	5-10 years	Yes	Yes
11	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	54		Главная акушерка	Городской перинатальный цен	Practicing	Midwife	11+ years	Yes	Yes
12	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Female	53		Председатль национальная ассоциация ак	НААРТ	Practicing	Midwife	11+ years	Yes	Yes
13	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Female	58		Декан	Европейский мультидисципли	Doctor		11+ years	Yes	Yes
14	2025-09-09	2025-09-09	Yes	Post	Tajikistan										
15	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	22		Семейная медсестра	ЦОВП ЧР ГСВ 1	Nurse		1 year	Yes	Yes
16	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Female	56		Замнач РМО по родовспоможению	Районное мед объединение Б	Doctor		11+ years	Yes	Yes
17	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Female	51		Неонатология	ЦРПКМР	Doctor		11+ years	Yes	Yes
18	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Female	52		Преподаватель	РЦПКиССМиФ	Others	Преподаватель	11+ years	Yes	Yes
19	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Female	57		Акушеока	Р. П. Ц.	Practicing	Midwife	11+ years	Yes	Yes
20	2025-09-09	2025-09-09	Yes	Pre	Kyrgyzstar	Female	50		Заведующая кафедрой	Кыргизский государственный	Doctor		11+ years	Yes	Yes
21	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Female	54		Внештатный главный акушер, гинеколог Ре	Таджикский НИИ акушерства,	Doctor		11+ years	Yes	Yes
22	2025-09-09	2025-09-09	Yes	Pre	Tajikistan	Female	55		Директор	НЦРЗ	Doctor		11+ years	Yes	Yes
23	2025-09-09	2025-09-09	Yes	Pre	Uzbekista	Male	46		Заместитель директора по подготовке кад	Республиканский специализи	Doctor		5-10 years	Yes	Yes
24	2025-09-12	2025-09-12	Yes	Post	Kyrgyzstan										
25	2025-09-12	2025-09-12	Yes	Post	Uzbekistan										
26	2025-09-12	2025-09-12	Yes	Post	Kyrgyzstan										
27	2025-09-12	2025-09-12	Yes	Post	Uzbekistan										
28	2025-09-12	2025-09-12	Yes	Post	Uzbekistan										

2nd Iteration – Designing 3 Power BI-Flows

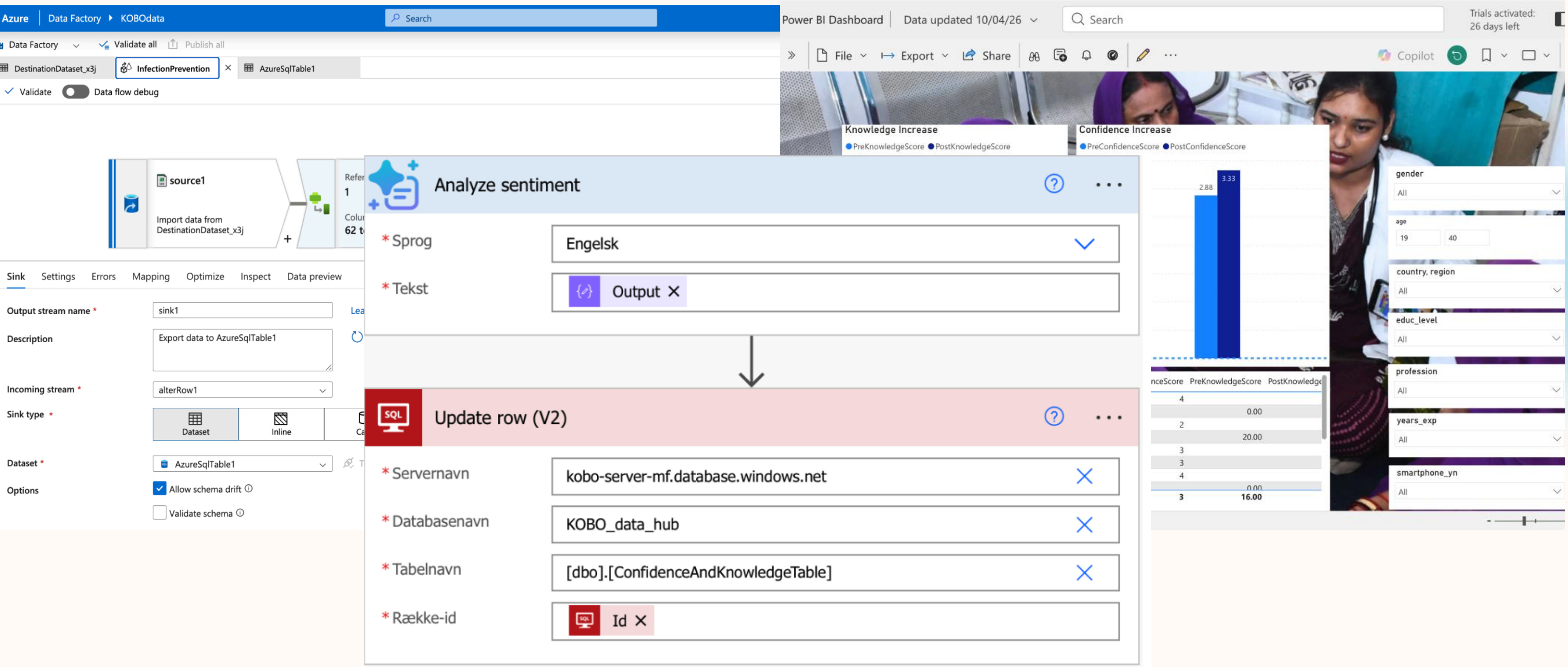


2nd Iteration - The 'Favourite' Flow





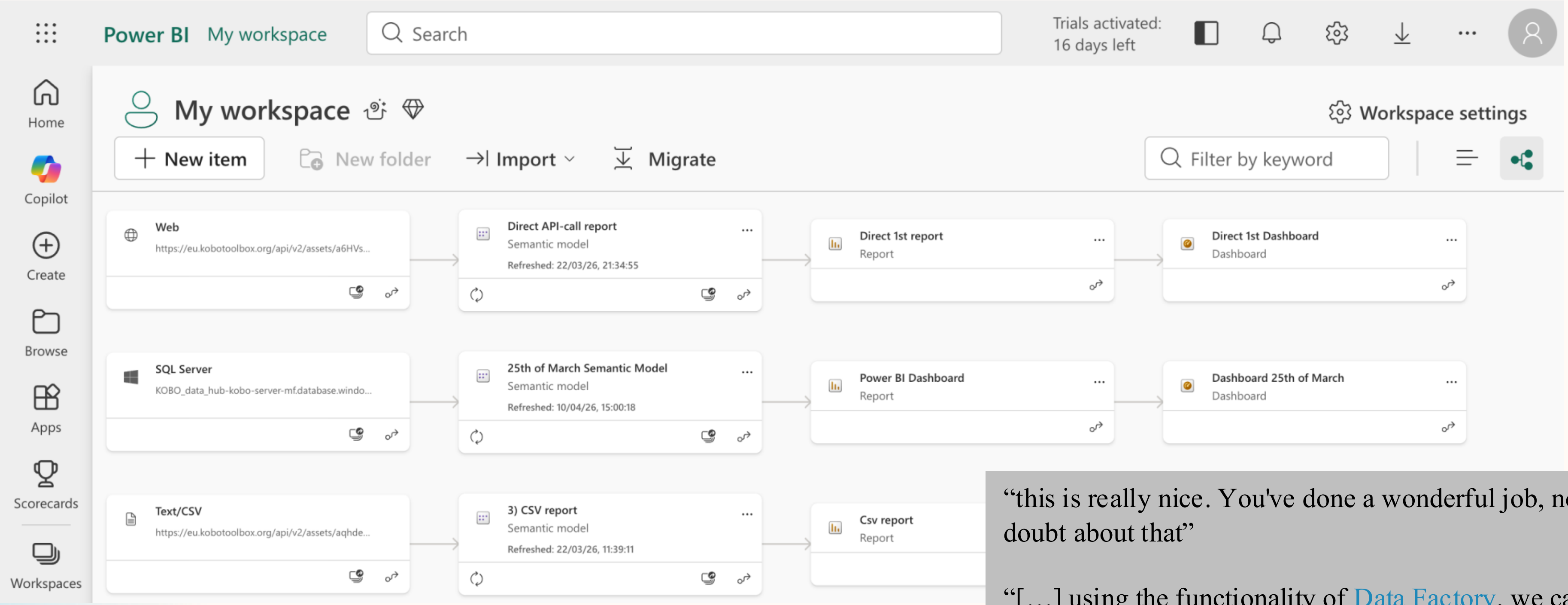
3rd Iteration – 'Proof-of-concept'



07

Demonstration and Evaluation

1st Demonstration and Evaluation – 'Formative' (technical)



"this is really nice. You've done a wonderful job, no doubt about that"

"[…] using the functionality of Data Factory, we can actually pull the data from Kobo and put it into the SQL Server DB, that would really help"

2nd Demonstration and Evaluation – 'Summative' (organizational/social)

Criteria	Description	Answer
Common language - making process automation trade-offs clear	Does it clarify the potentials and challenges of implementing a process automation solution concretely?	“if we can...quite clearly map out the advantages, the potential analyses and value addition [...]. I think it will make a great conversation to have with people who make decisions...
Facilitation of concrete discussion on standardization and flexibility	Does it stimulate concrete organizational discussion and dialogue around this topic?	I’m not sure if KOBO at this stage can handle something like that [...]. If we cannot do it, then what? Of course, we all want to do it because it’s easier and it makes sense logically, but what if we are not able to?”
Stimulate a discussion around what counts as “impact”	Does it facilitate reflections regarding what counts as impact – and letting this guide the data-collection and data-analysis?	“there's a desire to make progress as an organization and reaching a consensus-based conceptualization of impact that we can then try to provide data for.”
Inform future digital strategy	Does it illustrate tensions relevant to for their future digital strategy?	See above

Drawbacks

08

Design Knowledge (Findings)

What did we find?

Answering our research question through Design Science Research: How can an artifact be designed to support impact reporting in an NPO, and what design considerations and principles can be derived from this?

1) Impact Reporting in Maternity Foundation appeared a process automation problem.

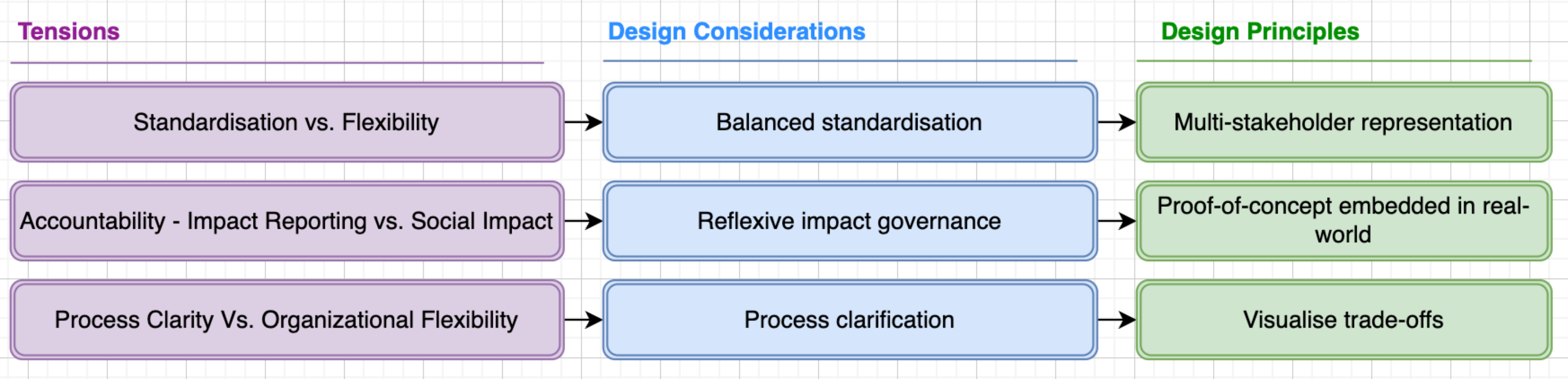
2) Through iterative DSR cycles we discovered it was a socio-organizational problem.

3) The artifact evolved from a techincal impact reporting automation to an artefact that also facilitates common language.

Key Insight:
Automating impact reporting is not primairly a technical challenge, but a challenge of navigating competing organizational priorities which can be addresed by facillitating cross-depармental discussion and contribute to establishing a common language

Design Knowledge: Making Tensions Negotiable

*The role of the **artifact** is not to eliminate tensions, but to make them visible and negotiable through **common language**.*



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Reflections

Key Reflections

From Common Language to Digital Strategy

- Radu et al. (2014) argues that common language supports digital transformation
- Our study identifies tensions that should be negotiated to establish common language in NPOs.
- How does common language ultimately translate into a concrete digital strategy?

The Resource Challenge

- Facilitating common language requires stakeholder involvement & dialogue
- NPOs often face resource constraints
- How do we balance participation and prototyping with limited resources?

Balancing Operational Performance and Social Mission

- Increasing demands for accountability drive quantification and standardization.
- However, excessive focus on measurable outcomes risks overlooking social value creation.
- How and who should determine this balance?

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MF Master Form (Knowledge)

Do you agree to participate?

- ◯ Yes
- ◯ No

Is this a Pre-training or Post-training assessment? (Pre-training - before the training and Post-training - after the training)

- ◯ Pre
- ◯ Post

Introduction and Background

Thank you for taking the time to complete this registration form. This form collects basic details to help us organize and improve the training. Your responses are confidential and used only for training and analysis purposes.

Please select your country

- ◯ country

Please select your region

Please enter your full name without titles (no Dr., Mr., Ms., Mrs., etc.)

Please enter your age

Please answer with a number. Example: 3

10

Post-submission work

Survey – Question 1

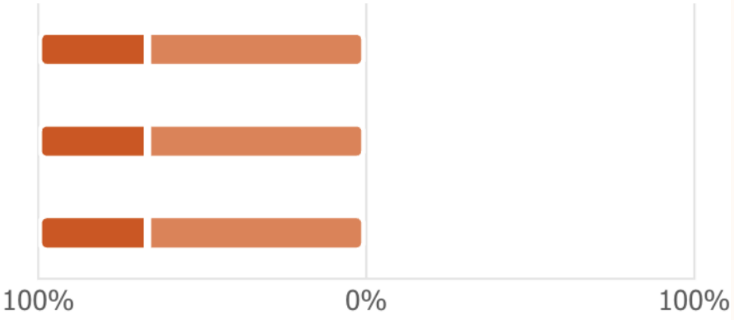
1. Questions

● Strongly agree ● Agree ● Neither agree nor disagree ● Disagree ● Strongly disagree

Have you gotten a better understanding of the challenges related to automating impact reporting?

Have you gotten a better understanding of the possibilities related to automating impact reporting?

Did the artefact help you reflect on the organisational value and strategy of Maternity Foundaton?



Survey – Question 2

Did the KOBO-dashboard and our presentation change your perspective on automating the impact reporting process?

"Yes, it was a very insightful and stimulating discussion. It highlighted how our existing data can be better connected and leveraged to automate the impact reporting process going forward"

"It generated interesting questions for our team to reflect on, particularly in terms of the pro's and con's of automatic impact reporting. It is clearly a complicated process to implement, but you shared some very valuable recommendations which will help us as we navigate the next level of our IT platform, our customer experience, and how we can draw better and more useful information from our data in terms of our impact."

"Yes, it did. It gave us a viable solution that we can use in our workflow."

Survey – Question 3

Do you have overall feed-back related to our presentation and KOBO-dashboard from 29th of April?

"The whole session was very engaging and insightful. It clearly demonstrated the potential of leveraging our existing data more effectively and sparked valuable discussions on how we can move towards automating the impact reporting process. The idea of connecting KOBO data with app data in the future is particularly promising, and it would be great to further plan how this integration can be achieved"

"I really appreciated that – in both the presentations I attended with you – you understood your audience very well. I work in fundraising, so my expertise is not in the technical side, but my interest lies in how we can better measure and report on our impact to funders. You asked for and our questions and diverse perspectives in the beginning, and acknowledged and addressed the differences in our MF internal group throughout the process. In that way, there was value for all of us in what you did, because each point of view was incorporated from the beginning. That's not an easy thing to do, and I applaud you for managing that so well. Thank you for your time and hard work."

"It was well-presented and featured good discussions around the benefits, challenges, and limitations. We look forward to adopting the solution provided by the team."

Thank you